SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-033: Last Login Tracking and Display for Student Accounts

Field	Details
Change Request ID	CR-033
Change Request Title	Last Login Tracking and Display for Student Accounts
Student Name	M M Amal
Student ID	JF/ICT/24/15
Implementation Status	Completed
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators could view student account and profile information, but the student list did not show when each student last logged in. The change was required to record the date and time of a student's most recent successful login and display it in the administrator's student table. This helps administrators understand recent account activity and identify students who have never logged in.

1.2 Existing Behaviour

Before implementation, the login process validated the username and password and created a JWT token after successful authentication. However, the system did not store or display the student's last login time.

The administrator's student table displayed the registration number, name, email, phone number, and available actions. Administrators could not see the date and time of a student's most recent successful login.

1.3 Requested Change

The system was required to:

- Add a nullable last_login timestamp to the users table.
- Update the timestamp only after successful username and password verification.
- Store the current database date and time using NOW().
- Include last_login in student records returned by the existing student API.
- Display a readable **Last Login** value in the administrator's student table.
- Display a suitable value for accounts that have never logged in.
- Preserve existing authentication, authorization, and student Add, Edit, and Delete functionality.

2. Implementation

2.1 Implementation Summary

The CR-033 requirements were implemented across the database, backend authentication flow, student data retrieval, and administrator dashboard.

A nullable last_login column was added to the users table. After a student's password is successfully verified, the backend updates that user's timestamp using the current database time. Student records returned by the existing student API now include the last_login value. The administrator dashboard displays it in a new **Last Login** column using a readable date and time format. Accounts without a recorded login display **Never logged in**.

The existing login response, JWT authentication flow, authorization controls, and student management actions were preserved.

2.2 Affected Files

File	Change	Purpose
database/schema.sql	Modified	Adds the nullable last_login column to the users table.
database/migrations/CR033_add_last_login.sql	Added	Adds the last_login column to an existing database.
backend/src/models/userModel.js	Modified	Adds the function to update a user's last login time.
backend/src/controllers/authController.js	Modified	Updates the timestamp after successful password verification.
backend/src/models/studentModel.js	Modified	Returns last_login with student records.
frontend/src/pages/DashboardPage.jsx	Modified	Displays and formats the Last Login value in the administrator's student table.

2.3 Technical Changes

Database: A nullable DATETIME column named last_login was added to the users table. Its initial value is NULL, allowing the application to identify accounts that have not logged in.

Login tracking: The backend updates last_login only after the submitted password is successfully verified. The update uses the database's current date and time. Failed login attempts do not update the timestamp.

Student data retrieval: The existing student queries retrieve last_login by joining student records with their related user records. No new API endpoint was required.

Administrator dashboard: A Last Login column was added to the student table. The timestamp is formatted for readability, and accounts without a valid recorded timestamp display Never logged in.

Existing functionality: Student Add, Edit, and Delete actions continue to use the existing student identity. The existing authentication and authorization flow remains in place.

2.4 Implementation Logic

1. Add the nullable last_login column to the users table.
2. Validate the submitted username and password using the existing login process.
3. After successful password verification, update that user's last_login value using the current database time.
4. Continue the existing login flow and return the existing authentication response.
5. Include last_login in student records retrieved through the existing student API.
6. Display the value in the administrator's student table using a readable date and time format.
7. Display **Never logged in** when no valid timestamp exists.
8. Preserve existing authentication, authorization, and student Add, Edit, and Delete behaviour.

2.5 Error and Data Handling

The last_login column allows a NULL value so that accounts without a previous successful login can be handled. The dashboard displays **Never logged in** when the timestamp is missing or invalid.

A failed login does not update the timestamp. Existing login error handling and authentication bhaviour remain in place.

3. Testing & Verification

3.1 Testing Approach

The implementation was manually verified in the development environment.

Testing types:

- Functional testing
- Positive testing
- Validation and error handling
- User interface testing
- Security behaviour verification
- Regression testing for existing student management functions

Environment: Development
Browser: Chrome
Test Role: Student and Administrator

3.2 Test Cases

Test Case	Expected Result	Actual Result	Status
Successful student login	The student's last_login value is updated with the current date and time.	Timestamp updated successfully.	Pass
Login with an incorrect password	Login is rejected and last_login remains unchanged.	Login rejected; timestamp remained unchanged.	Pass
Load the administrator's student list	Student records include the last_login value.	Last Login values were returned and displayed.	Pass
View a student account with no previous login	The dashboard displays "Never logged in."	"Never logged in" was displayed.	Pass
Edit a student record	The existing Edit action continues to work for the correct student.	Existing Edit behaviour worked correctly.	Pass
Delete a student record	The existing Delete action continues to work for the correct student.	Existing Delete behaviour worked correctly.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	6
Passed	6
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	A successful login records the student's latest login date and time.	Successful login test	Met
AC-002	Incorrect login credentials do not update the last login timestamp.	Incorrect password test	Met
AC-003	The administrator's student table displays a Last Login column.	Dashboard verification	Met
AC-004	Accounts without a previous login display an appropriate value.	Never logged in test	Met
AC-005	Existing student management operations continue to work.	Edit and Delete regression tests	Met
AC-006	Existing authentication and authorization behaviour remains unchanged.	Login and access behaviour verification	Met

5. Evidence

Evidence 01 – Last Login Display

Related: AC-003, AC-004

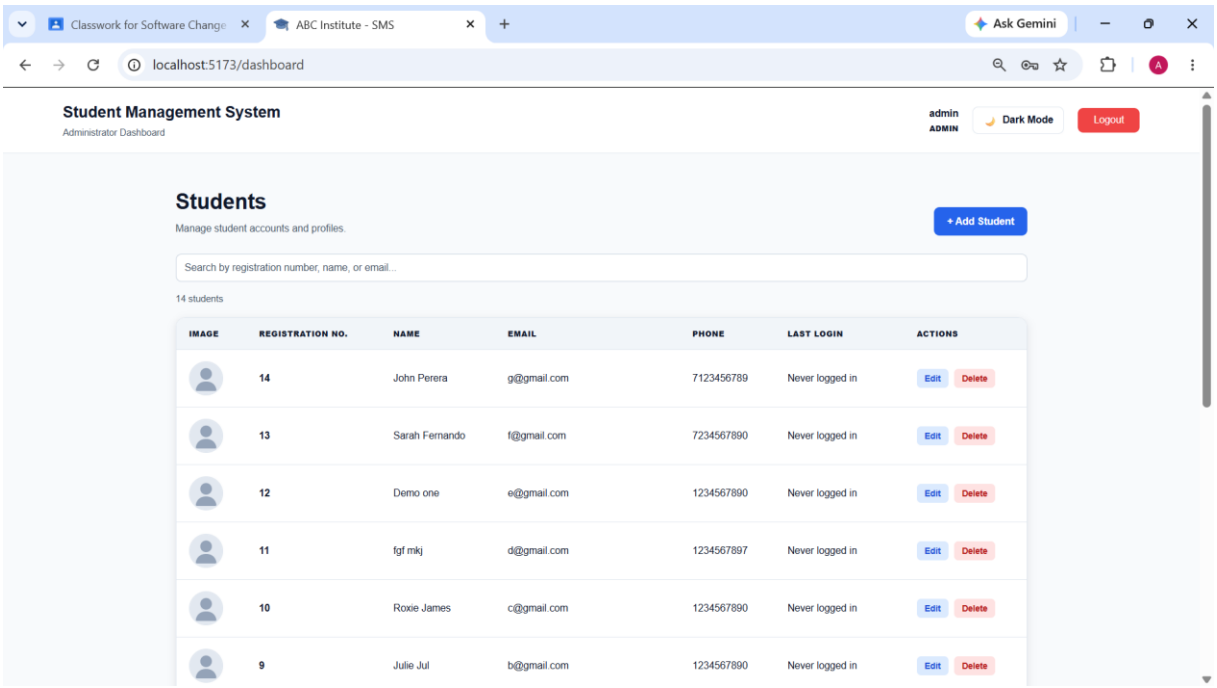
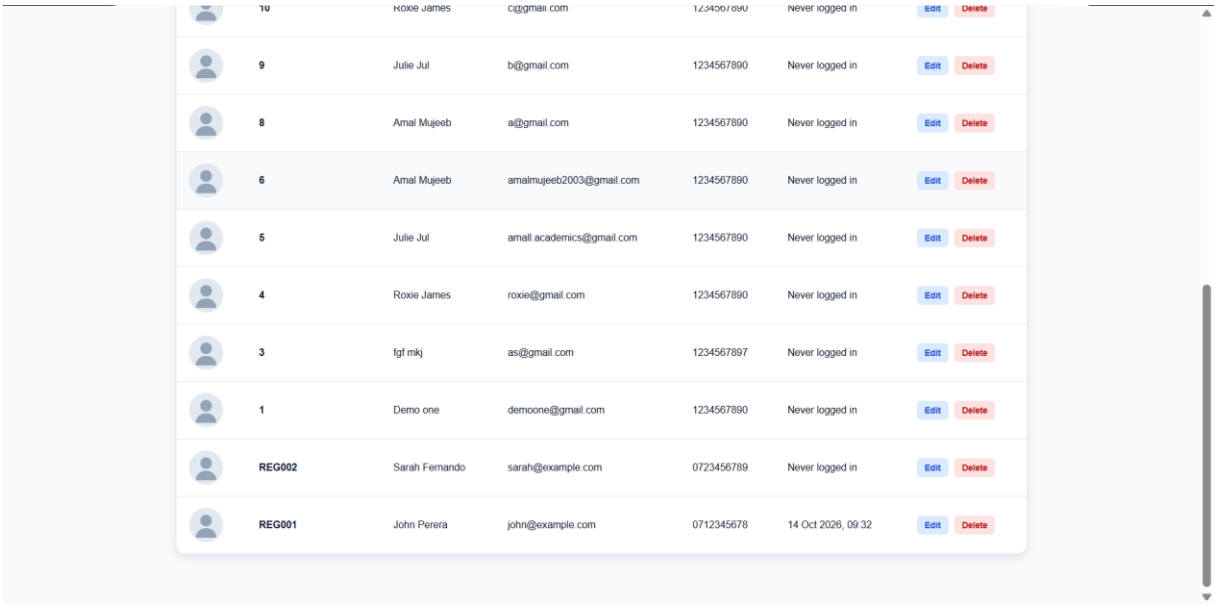


Figure 1 Administrator dashboard showing the Last Login column with formatted timestamps and the "Never logged in" value where applicable.

Evidence 02 – Successful Login and Timestamp Update

Related Requirements: AC-001, AC-002



10	Roxie James	c@gmail.com	1234567890	Never logged in	Edit	Delete
9	Julie Jul	b@gmail.com	1234567890	Never logged in	Edit	Delete
8	Amal Mujeeb	a@gmail.com	1234567890	Never logged in	Edit	Delete
6	Amal Mujeeb	amalmujeeb2003@gmail.com	1234567890	Never logged in	Edit	Delete
5	Julie Jul	amall.academics@gmail.com	1234567890	Never logged in	Edit	Delete
4	Roxie James	roxie@gmail.com	1234567890	Never logged in	Edit	Delete
3	fgf mkj	as@gmail.com	1234567897	Never logged in	Edit	Delete
1	Demo one	demoone@gmail.com	1234567890	Never logged in	Edit	Delete
REG002	Sarah Fernando	sarah@example.com	0723456789	Never logged in	Edit	Delete
REG001	John Perera	john@example.com	0712345678	14 Oct 2026, 09:32	Edit	Delete

Figure 2 Successful student login followed by the updated Last Login value. Do not include passwords or other sensitive authentication data.

6. Final Result

Item	Result
Requirements	Verified
Implementation	Implemented
Testing	Verified
Acceptance Criteria	Verified
Existing CRUD	Verified
Final Status	Verified

6.1 Final Summary

CR-033 was implemented to record the date and time of each student's most recent successful login and display it in the administrator's student table. The timestamp is updated only after successful password verification. Student records include the timestamp through the existing student API, and accounts without a login record display **Never logged in**.

Testing confirmed that successful logins update the timestamp, failed logins do not, and existing student Edit and Delete operations continue to work. All six test cases passed.

6.2 Issues / Limitations

No significant issues were identified during implementation and testing.

Student Name: M M Amal

Student ID: JF/ICT/24/15

Date: 13 October 2026

Signature: Amal